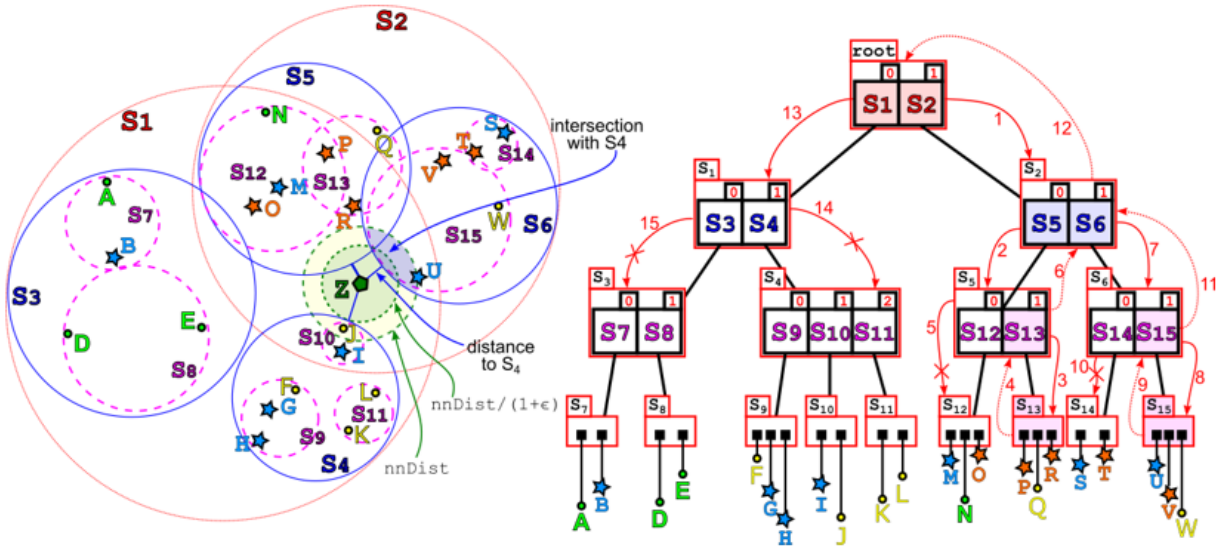


Figure 10.24 Approximated nearest neighbor search. The example is similar (almost identical) to the one in figures 10.20-21, to allow a close comparison. We show the verbose representation of the tree to make it clearer the path followed in traversal.
Node S_4 contains J, the true NN, bit is further away from the query point Z than S_5 and S_6, and outside the approximated query region (ϵ has been chosen ad-hoc, obviously, to cause this condition: in this example, ϵ~=0.25). Arrows are numbered in the order in which they are traversed.

In many domains we can be not just fine, but even happy, with approximate search. Take our example about the search through a dataset of images: can we really be sure that an exact match is better than an approximate one? If we were working with geographic coordinates, say on a map, then a factor ϵ difference can have dire consequences (at the very best, taking a longer route would be expensive, but it might get as bad as to safety issues). But when the task is finding